COMP_SCI 358
Intro to Parallel Computing
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Northwestern University

Parallelism in Real Life: How Pixar Renders Movies

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Pixar’s Animation Pipeline

Step 1: Modeling

Build 3D characters and environments

Step 2: Animation

Make characters move and act

Step 3: Lighting

Add virtual lights and mood

Step 4: Rendering

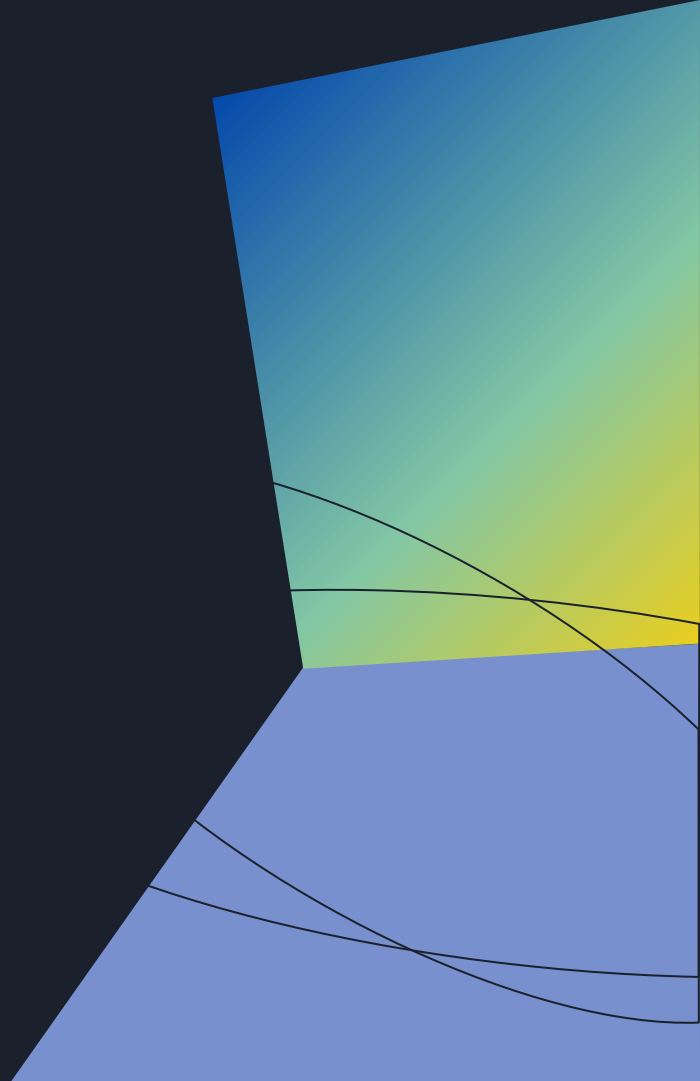
Convert scenes into 2D images

Goal

Final movie

What is Rendering?

- Rendering is the final step that turns 3D scenes into images
- Pixar Films: 24 frames/sec $\times$ 90 mins = 129,600 images
- Rendering is computationally expensive!



Why Is Rendering So Expensive?

- High resolution: millions of pixels per frame
- Complex lighting & materials (hair, fire, water, etc.)
- Long Render Times: Up to 90 hours/frame
- Cost of hardware, electricity, and time

Frame-Level Parallelism with HPC

- Each frame is rendered independently → embarrassingly parallel
- Pixar uses HPC clusters to process frames in parallel
- Example: Toy Story 4 ≈ 122,400 frames

Intra-Frame Parallelism

& Denoising in Path Tracing

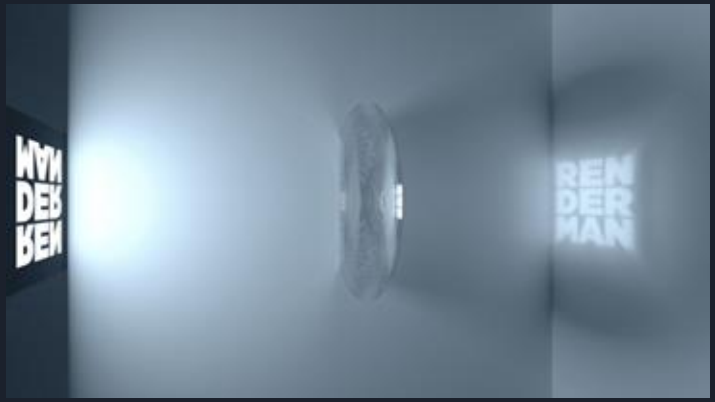
- Path tracing launches hundreds of rays per pixel, each computed in parallel
- Frame is divided into tiles, rendered concurrently across CPU/GPU threads
- Denoising filters applied per tile or image-wide using parallel passes
- Used in Pixar's first path-traced feature: *Finding Dory* (2016)

Pseudo Code

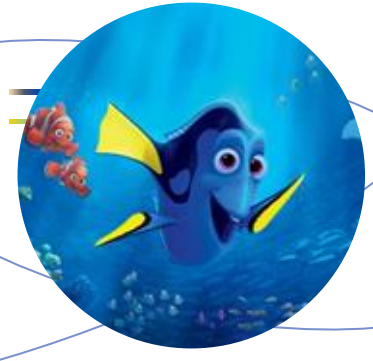
```
// c++
// launch one GPU thread per pixel

kernel void TraceFrame(Frame frame) {
    int x = get_global_id(0);
    int y = get_global_id(1);

    Ray ray = GeneratePrimaryRay(x, y);
    Color color = TracePath(ray);
    frame.setPixel(x, y, color);
}
```

Pixar's Tools: RenderMan in Action

- RenderMan is Pixar's proprietary renderer, optimized for parallel computing
- Supports advanced algorithms: VCM (Vertex Connection and Merging) and UPBP (Unified Points, Beams, and Paths)
- Used to simulate caustics, volumetric lighting, and complex reflections
- Handles multi-threaded photon maps to avoid read/write conflicts



Case Study



Finding Dory

Coco

Inside Out

Avg. Render Time per Frame	~ 7 hours	Reportedly up to 90 hours	11 - 20 hours
Parallel Complexity	Path tracing + denoising	Complex lighting, thousands of candles	Particle effects, glowing orbs

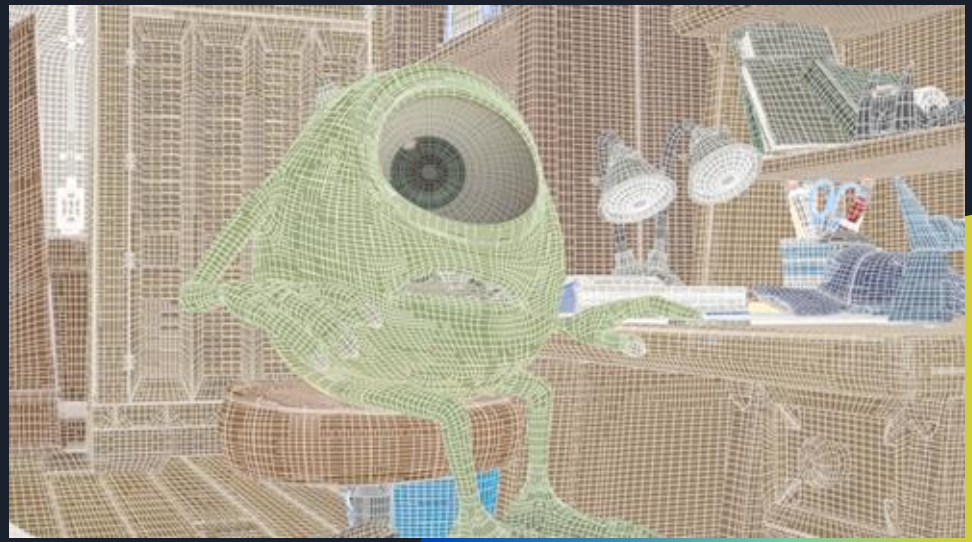
Why Parallelism Matters

- Enables creative freedom
  → fast iterations
- Keeps multi-year projects on schedule
- Makes photorealism scalable

From Code to Cinema

- Rendering = ideal parallel problem
- Pixar uses frame, tile, and ray-level parallelism
- Tools like RenderMan + HPC make it possible
- Parallel computing brings Pixar's stories to life

[Frame-Level] (HPC) → [Tile-Level] (Multithreading) → [Ray-Level] (GPU cores)



Sources

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